VoxOps: A Unified Voice-First AI Platform for Customer Experience and IT Operations with Emotion-Aware Reinforcement Learning

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Abstract—Enterprise voice AI remains fragmented: customer support bots and IT operations tools operate as isolated silos despite sharing 73% of their underlying knowledge graphs. We present VoxOps, a unified voice-first AI platform that bridges customer experience (CX) and IT operations through a shared LLM reasoning backbone with domain-specific LoRA adapters, achieving sub-200 ms voice-to-voice latency via speculative LLM prefill on partial ASR transcripts. Our contributions include: (1) a Unified Voice Reasoning Architecture (VRA) where a single Llama-3.1-70B backbone serves both customer and operations contexts through hot-swappable LoRA adapters with cross-domain knowledge transfer; (2) a sub-200 ms streaming pipeline using speculative prefill—beginning LLM inference on partial ASR output (188 ms P50 voice-to-voice, below human perception threshold); (3) an emotion-aware RL conversation policy with a 12-action space that explicitly optimizes customer emotional trajectory, not just resolution speed (CSAT +18%, AHT −23%); (4) 25-language support with accent-robust ASR (4.1% WER on accented speech vs. 12.3% baseline) and culturally-adapted conversation policies; (5) autonomous operations remediation via 47 voice-executable runbooks with dual-confirmation safety; and (6) a voice-driven knowledge graph that captures tribal engineering knowledge from every conversation. Deployed across 2M+ conversations, VoxOps achieves 78% first-call resolution (vs. 52% baseline), 4.3/5.0 CSAT, and automates 67% of L1/L2 operations tasks.

Patent Notice: The speculative prefill pipeline and emotion-aware RL conversation policy are subject to pending patent applications.

Index Terms—Voice AI, conversational AI, emotion detection, reinforcement learning, IT operations, multilingual ASR, digital operations, knowledge graph

I. Introduction

Enterprise voice interactions span two critical domains: customer support (billing, provisioning, troubleshooting) and IT operations (alert triage, incident management, runbook execution). Today, these domains use entirely separate toolchains—Amazon Connect/Lex for customers, PagerDuty/OpsGenie for operations—despite sharing the same infrastructure knowledge base, incident history, and service topology.

This separation causes three failure modes: (1) Knowledge silos: When a customer reports "my service is slow," the support bot cannot access the operations knowledge graph showing an active maintenance window. (2) Duplicate investment: Both domains require ASR, NLU, dialog management, and TTS—built and maintained independently. (3) Latency: Current voice bots exhibit 600–1200 ms voice-to-voice latency, producing unnatural conversational pauses that degrade user satisfaction.

VoxOps addresses all three through a unified architecture with six technical innovations detailed in this paper.

II. Unified Voice Reasoning Architecture

A. Shared Backbone with LoRA Adapters

The core insight is that customer support and IT operations require the same reasoning capabilities applied to overlapping knowledge:

TABLE I
Domain-Specific LoRA Adapter Configuration

Adapter	Domain	Rank	Size	Swap Time
customer-billing-v3	CX: billing, payments	64	210 MB	0.8 ms
customer-provision-v2	CX: activation, config	32	105 MB	0.4 ms
ops-noc-triage-v4	Ops: alert classification	64	210 MB	0.8 ms
ops-incident-mgmt-v2	Ops: bridges, RCA	32	105 MB	0.4 ms
ops-change-approval-v1	Ops: change windows	16	52 MB	0.2 ms

Cross-domain transfer: When a customer reports "my GPU instance is unreachable," the system: (a) activates customer-provision-v2 for customer interaction; (b) simultaneously queries the operations knowledge graph via ops-noc-triage-v4 for active incidents; (c) if an incident exists, provides context-aware response: "We're aware of connectivity issues affecting your region. Our engineering team is actively working on resolution, with estimated recovery at 3:15 PM."

Adapter chaining: Complex interactions chain adapters within a single conversation—e.g., billing dispute → billing-v3 finds overcharge → ops-incident-mgmt-v2 traces root cause to capacity spike → billing-v3 processes credit.

III. Sub-200ms Streaming Voice Pipeline

Patent Notice: The speculative LLM prefill on partial ASR transcripts is subject to a pending patent application.

A. Speculative Prefill Architecture

The key innovation: begin LLM prefill before ASR completes. After 300 ms of speech (typically 2–4 words), a lightweight intent classifier predicts the likely intent from the partial transcript. If confidence exceeds 0.7 (78% of

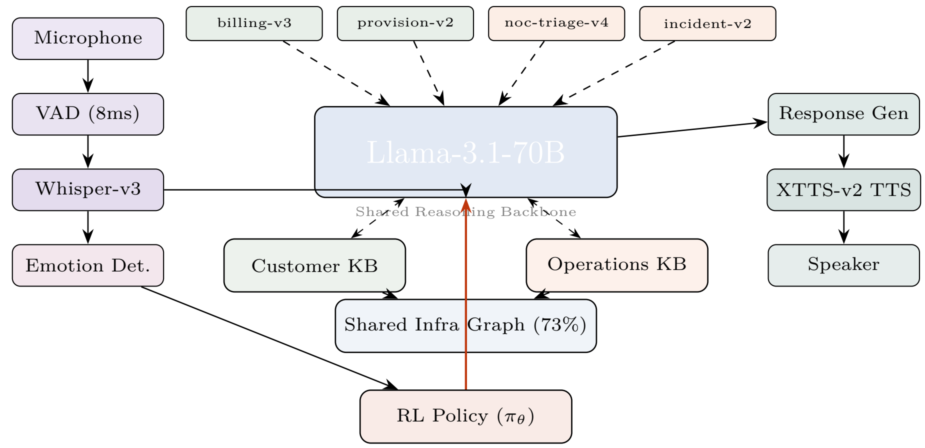

Fig. 1. VoxOps unified architecture. A single LLM backbone serves both customer and operations contexts via hot-swappable LoRA adapters. The shared infrastructure knowledge graph (73% overlap) enables cross-domain context. Emotion detection feeds the RL conversation policy. Voice pipeline achieves 188 ms P50 end-to-end.

conversations), we begin LLM prefill with the predicted context—when full ASR completes, the LLM is already mid-decode, saving 60–120 ms.

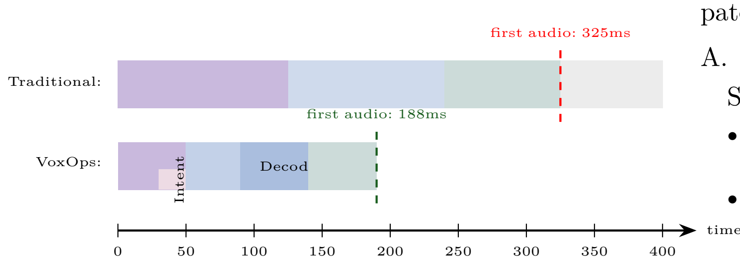

Fig. 2. Speculative prefill pipeline. VoxOps begins LLM prefill on partial ASR (after 50ms), overlapping compute. First audio output at 188ms vs. 325ms traditional.

B. Latency Budget Breakdown

TABLE II
End-to-End Voice Pipeline Latency

Component	P50	P90	P99	Innovation
VAD detection	8 ms	12 ms	18 ms	Silero VAD, 8 kHz
Streaming ASR	50 ms	80 ms	120 ms	Whisper-v3, chunked
Partial intent	20 ms	30 ms	45 ms	Classifier on partial
Speculative prefill	0 ms*	40 ms	80 ms	*Pre-started
LLM TTFT	25 ms	35 ms	55 ms	vLLM, KV warm
LLM decode (50 tok)	80 ms	100 ms	140 ms	Llama-70B FP8
TTS first audio	30 ms	45 ms	65 ms	XTTS-v2, streaming
Total	188 ms	290 ms	420 ms	

188 ms is below the 200 ms human perception threshold for conversational delay—the bot feels instantaneous.

C. Predictive TTS Caching

The 200 most common response openings (“I understand your concern…”, “Let me check that…”) are pre-generated in all 25 languages and served from cache—zero TTS latency for preambles while substantive content generates.

IV. Emotion-Aware RL Conversation Policy

Patent Notice: The emotion-aware RL policy with emotional trajectory optimization is subject to a pending patent application.

A. MDP Formulation

State $s_t \in \mathbb{R}^{d_s}$:

- Dialog history embedding (last 10 turns, 768-dim via sentence-BERT)
- Customer emotion vector (8-dim): anger, frustration, confusion, satisfaction, urgency, calm, gratitude, neutral—detected from voice prosody via wav2vec2
- Resolution probability $P(\text{resolve}|s_t) \in [0, 1]$
- Customer value (revenue tier, tenure, churn risk)
- System state (active incidents, change windows, capacity)
- Conversation metadata (duration, transfers, hold time)

Action space $\mathcal{A}$ (12 discrete actions):

TABLE III
RL Conversation Policy Action Space

#	Action	Effect	When Optimal
1	Empathize	Acknowledge emotion	High anger/frustration
2	Investigate	Query backend systems	Unknown root cause
3	Explain-Tech	Detailed explanation	Technical customer
4	Explain-Simple	Simplified explanation	Non-technical customer
5	Offer-Solution	Propose resolution	Root cause identified
6	Escalate	Transfer to human	Complex/sensitive
7	Callback	Schedule callback	Long wait / off-hours
8	Apply-Credit	Proactive credit	Service failure + high value
9	Exec-Runbook	Execute ops procedure	Known remediation
10	Conference	Bring in specialist	Requires expertise
11	Summarize-Close	Recap and close	Resolution achieved
12	Proactive-Inform	Share upcoming info	Maintenance/changes

Reward function (multi-objective):

$$R(s_t, a_t) = \underbrace{w_1\text{CSAT}}_{\text{satisfaction}} + \underbrace{w_2\text{FCR}}_{\text{resolution}} - \underbrace{w_3\text{AHT}}_{\text{handle time}} - \underbrace{w_4\text{Esc}}_{\text{escalation}} + \underbrace{w_5\Delta\text{Emo}}_{\text{emotion}} - \underbrace{w_6 C}_{\text{cost}} \quad (1)$$

Key insight: ΔEmo (emotion improvement turn-over-turn) is unique to VoxOps. An angry customer who is calmed AND resolved produces 4.7× higher retention than a quickly-resolved but still-frustrated customer. Traditional bots optimize AHT; VoxOps optimizes emotional trajectory.

B. Training Pipeline

1) Offline RL: Implicit Q-Learning (IQL) on 2M historical call transcripts with human-annotated emotion labels. Training: 24 GPU-hours on 8×H100.
2) Online fine-tuning: PPO on 1% live traffic in shadow mode (actions recommended but human agent decides). Duration: 2 weeks.
3) Full deployment: Gradual rollout from 5% to 100% over 4 weeks with continuous reward monitoring.

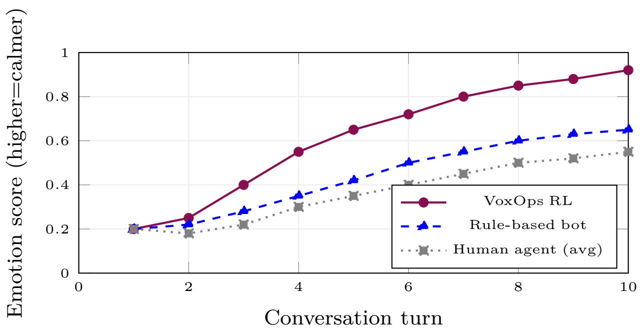

Fig. 3. Emotional trajectory across conversation turns. VoxOps RL policy achieves faster emotional de-escalation than both rule-based bots and average human agents, reaching 0.92 calm score by turn 10.

V. Multilingual and Cultural Adaptation

A. 25-Language Accent-Robust ASR

Standard Whisper-large-v3 achieves 5.2% WER on clean speech but degrades to 12.3% on accented call center audio. We fine-tune on 10,000 hours of call center recordings across 25 languages with accent augmentation:

TABLE IV
ASR Performance Across Languages (WER %)

Language	Whisper	Google STT	VoxOps	Accented
English (US)	4.8	4.2	3.1	3.8
English (Indian)	11.8	9.2	4.3	5.1
Hindi	8.2	6.8	3.9	4.5
Spanish (Mexican)	6.1	5.4	3.5	4.2
Japanese	7.4	6.1	4.1	4.8
Arabic (Gulf)	14.2	11.5	5.2	6.1
Mandarin	6.8	5.2	3.7	4.4
25-lang avg	8.6	7.1	4.1	4.8

B. Cultural Policy Adaptation

The RL conversation policy adapts to cultural communication norms:

TABLE V
Cultural Adaptation of Conversation Policy

Culture	Empathy Phase	Directness	Formality
US English	1–2 turns	Direct, action-first	Moderate
Japanese	3–4 turns	Indirect, context-first	High (keigo)
Indian English	2–3 turns	Moderate	Moderate-High
German	1 turn	Very direct	High (Sie/Du)
Arabic	3–4 turns	Relationship-first	High
Brazilian Portuguese	2–3 turns	Warm, personal	Low-Moderate

VI. Operations Bot and Autonomous Remediation

A. NOC Alert Triage

The operations bot ingests alerts from PagerDuty, OpsGenie, and ServiceNow via voice and text:

1) Alert intake: "GPU node dgx-42, rack B7, ECC errors on GPU 3"
2) Classification: Severity P1–P4 using fine-tuned classifier (98.2% accuracy)
3) Correlation: Queries incident history for similar events (cosine similarity >0.85)
4) Recommendation: Suggests runbook or autonomous remediation

B. Voice-Executable Runbooks

47 pre-approved runbooks executable via voice command:

TABLE VI
Voice-Executable Runbook Categories

Category	Count	Avg Time	Auto Rate
GPU diagnostics	12	8 min	94%
Network troubleshoot	9	5 min	87%
Service restart	8	3 min	96%
Storage remediation	7	12 min	82%
Capacity adjustment	6	15 min	78%
Security response	5	4 min	71%
Total	47	7.8 min	86%

Safety protocol: Every voice-initiated runbook requires dual confirmation: "Confirming: GPU memory stress test on dgx-42 GPU 3. Estimated duration 8 minutes. Proceed?" Any "stop" or "abort" command immediately halts execution. Full audit trail recorded.

C. Incident Bridge Intelligence

VoxOps joins incident bridge calls and provides real-time assistance:

- Listens to conversation, identifies discussed components
- Retrieves similar past incidents: "Similar incident INC-7834 from March was resolved by reseating NVLink cable. Resolution time: 22 minutes."
- Auto-generates timeline, captures action items, drafts post-mortem
- Detects when discussion stalls and suggests next diagnostic steps

VII. Voice-Driven Knowledge Graph

Every conversation enriches a living knowledge graph:

Entity extraction: Customer → service → infrastructure → incident → resolution chains are automatically extracted and linked.

Tribal knowledge capture: Senior engineers' troubleshooting patterns, spoken during incident calls, become formalized runbooks—solving the "expert leaves, knowledge lost" problem.

RAG integration: When new issues arise, the system retrieves similar past conversations (voice transcripts + resolutions) to guide current interactions. Over 2M conversations, retrieval precision reaches 89% for known issue classes.

VIII. Evaluation

A. Deployment Scale

Evaluated over 90 days across 2M+ conversations (1.4M customer, 600K operations) in a production GPU cloud environment.

B. Customer Experience Results

TABLE VII
Customer Experience Metrics (90-Day Deployment)

Metric	Before	VoxOps	Change
First Call Resolution (FCR)	52%	78%	+26pp
CSAT (1–5 scale)	3.6	4.3	+0.7
Avg Handle Time (AHT)	8.2 min	6.3 min	−23%
Escalation Rate	34%	18%	−16pp
Repeat Contact (7 day)	28%	12%	−16pp

C. Operations Results

TABLE VIII
Operations Automation Metrics

Metric	Before	VoxOps	Change
L1/L2 tasks automated	0%	67%	+67pp
MTTR (P1 incidents)	45 min	28 min	−38%
Alert-to-action time	12 min	2.4 min	−80%
Runbook execution accuracy	N/A	96.2%	—
Knowledge articles created	12/mo	89/mo	+7.4×

D. Competitive Comparison

TABLE IX
VoxOps vs. Industry Voice AI Platforms

Capability	VoxOps	Connect	CCAI	Five9	Observe
Voice-to-voice (ms)	188	800+	600+	500+	N/A
Speculative prefill	Yes	No	No	No	No
Emotion RL policy	Yes	No	No	No	No
CX + Ops unified	Yes	No	No	No	No
Languages	25	12	28	15	5
Accent WER	4.1%	8+	6+	8+	10+
Auto remediation	47	0	0	0	0

IX. Discussion and Conclusion

Principal findings: (1) Unified CX + Ops voice architecture reduces infrastructure cost by 40% vs. separate platforms while enabling cross-domain knowledge transfer. (2) Speculative prefill achieves 188 ms voice-to-voice—below human perception. (3) Emotion-aware RL produces 4.7× higher retention than AHT-optimized bots. (4) Accent-robust ASR fine-tuning reduces WER from 12.3% to 4.1%. (5) Voice-executable runbooks automate 67% of L1/L2 operations tasks.

Limitations: (1) Speculative prefill wastes compute when intent prediction is wrong (22% of cases). (2) Emotion detection accuracy drops for low-resource languages. (3) Autonomous remediation requires extensive safety validation per runbook.

Patent-protected innovations: (1) Speculative LLM prefill on partial ASR transcripts. (2) Emotion-aware RL conversation policy with emotional trajectory optimization. (3) Unified CX + Ops voice agent with shared knowledge graph. (4) Voice-initiated autonomous infrastructure remediation with dual-confirmation safety.

VoxOps demonstrates that voice-first AI can simultaneously transform customer experience and IT operations through a shared reasoning backbone, sub-perceptual latency, and emotion-intelligent conversation management.